SPL-1 | MIDTERM PROGRESS PRESENTATION

Short-Term Traffic Congestion Prediction and Alternate Route Recommendation System

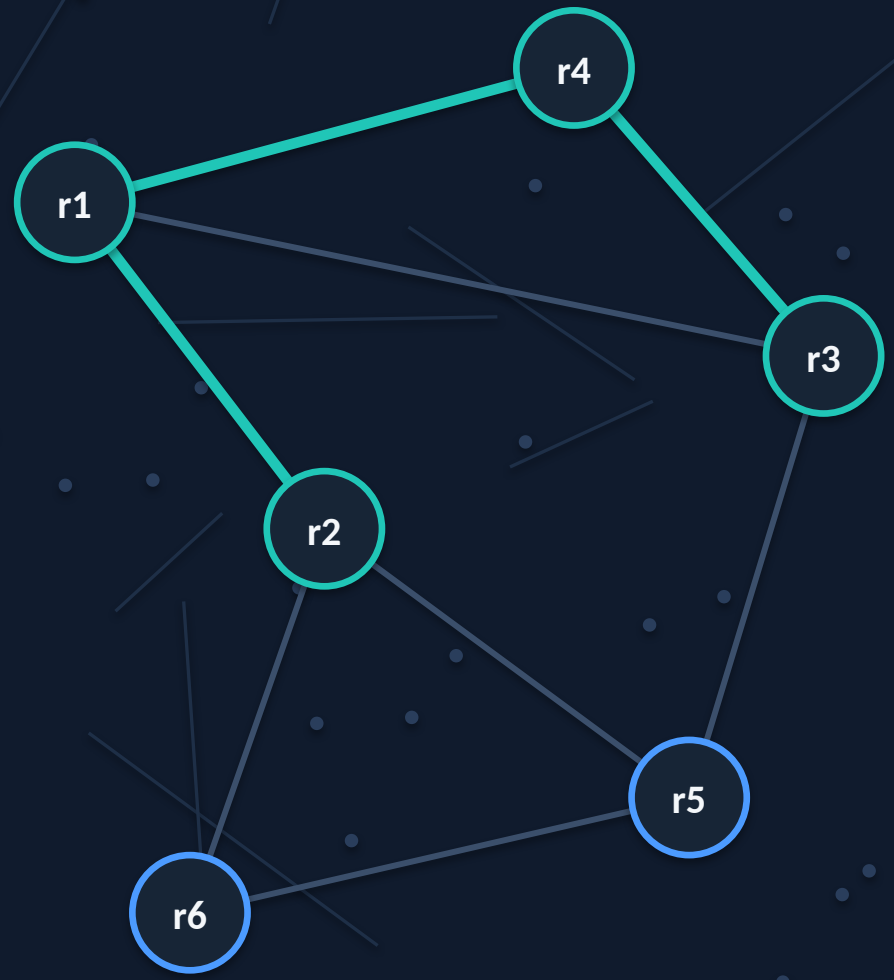
Predicting speed 15 minutes ahead, then choosing the fastest estimated route

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Why a route's travel time is hard to trust

ROUTE A | DIRECT r2 to r3 — slower forecast

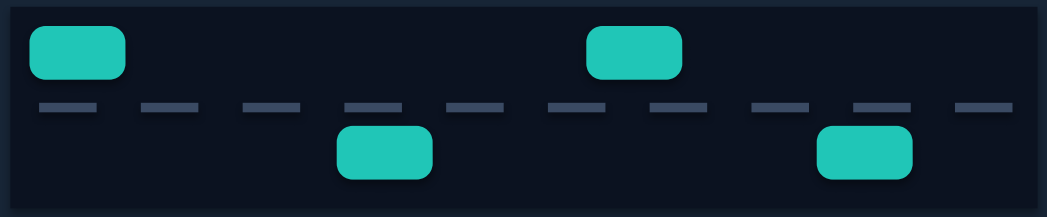


Forecast at r2: 33.0 mph

6.84 min estimated time, direct leg

vs

ROUTE B | VIA r1 r2, r1, r3 — higher forecast



Forecast at r1: 54.3 mph

4.54 min estimated time via r1

Example: committed 2-day sample, forecast from row 489. Model estimates, not measured trips.

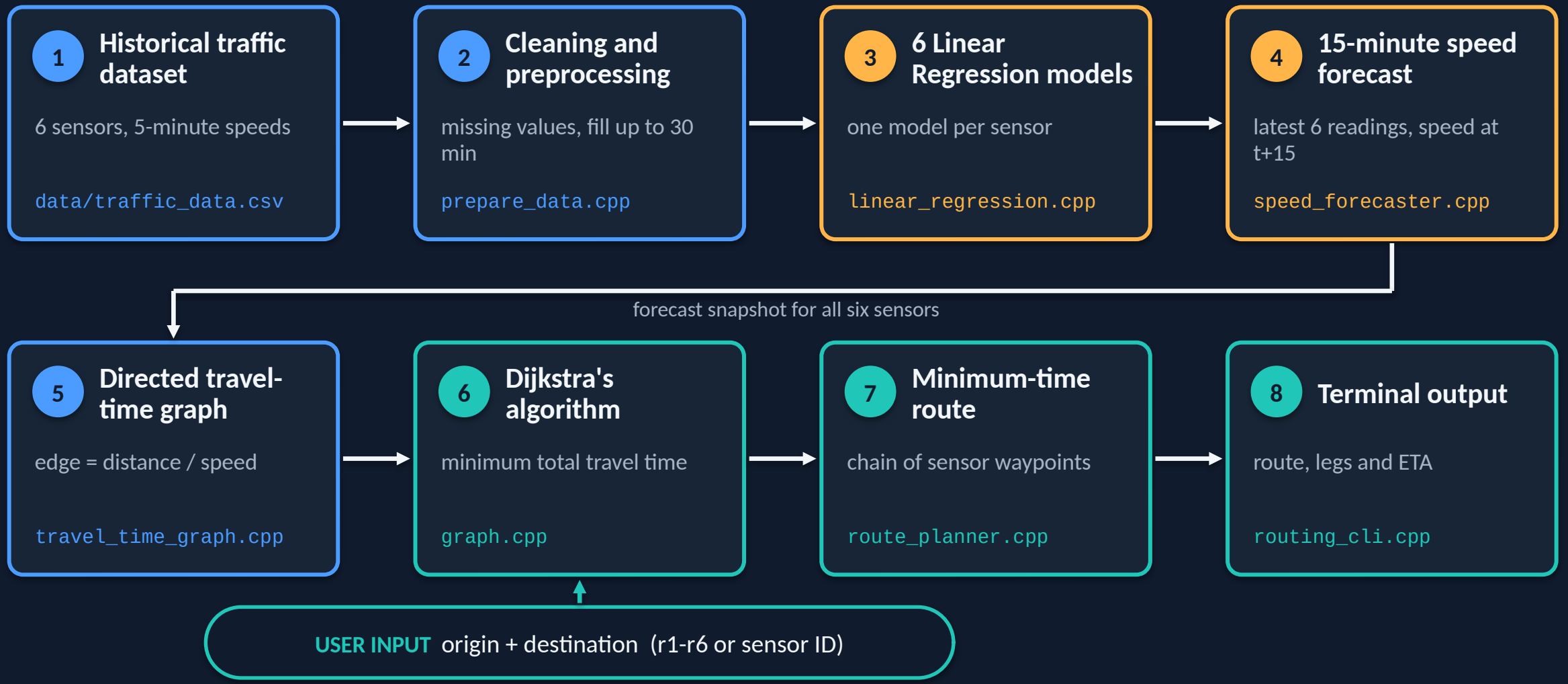
THE PROBLEM

Travel time depends on traffic that keeps changing. Choosing a route from current speed alone ignores the next 15 minutes.

THE IDEA

Forecast speed 15 minutes ahead, convert it to travel time, then pick the route with the lowest estimate.

System architecture: from history to route



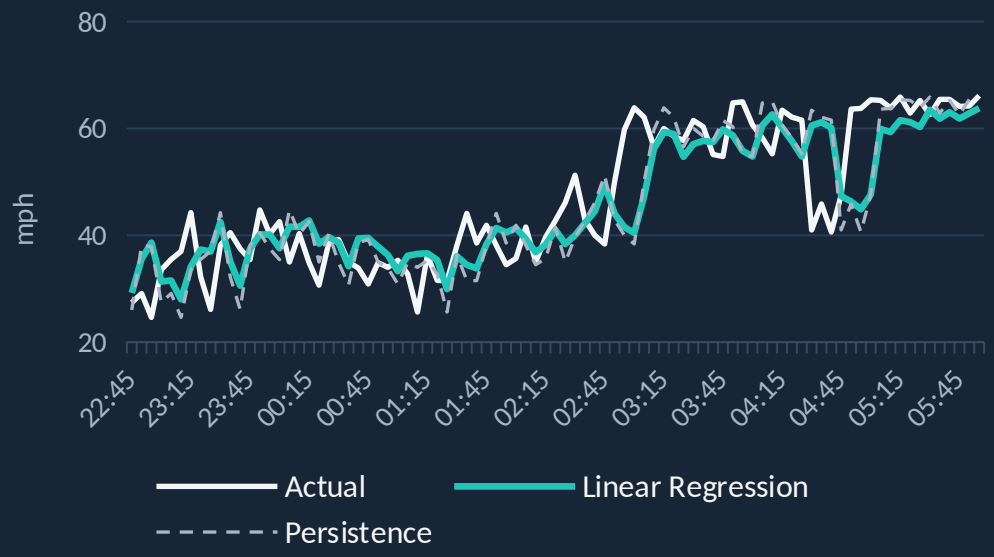
Data, preprocessing and Linear Regression

6	5 min	34,272
traffic sensors	observation interval	observations (full dataset)

Full dataset is local (not in GitHub): Mar 1 to Jun 27, 2012. GitHub sample: 576 rows, 2 days.

1 Clean	zero, negative and NaN readings become missing
2 Fill gaps	carry last reading forward, at most 6 steps
3 Lag features	latest 6 readings (30 min) per sensor
4 Chronological split	70% train, 15% validation, 15% test
5 Train 6 models	standardize, then gradient descent
6 Evaluate	MAE and RMSE vs persistence baseline

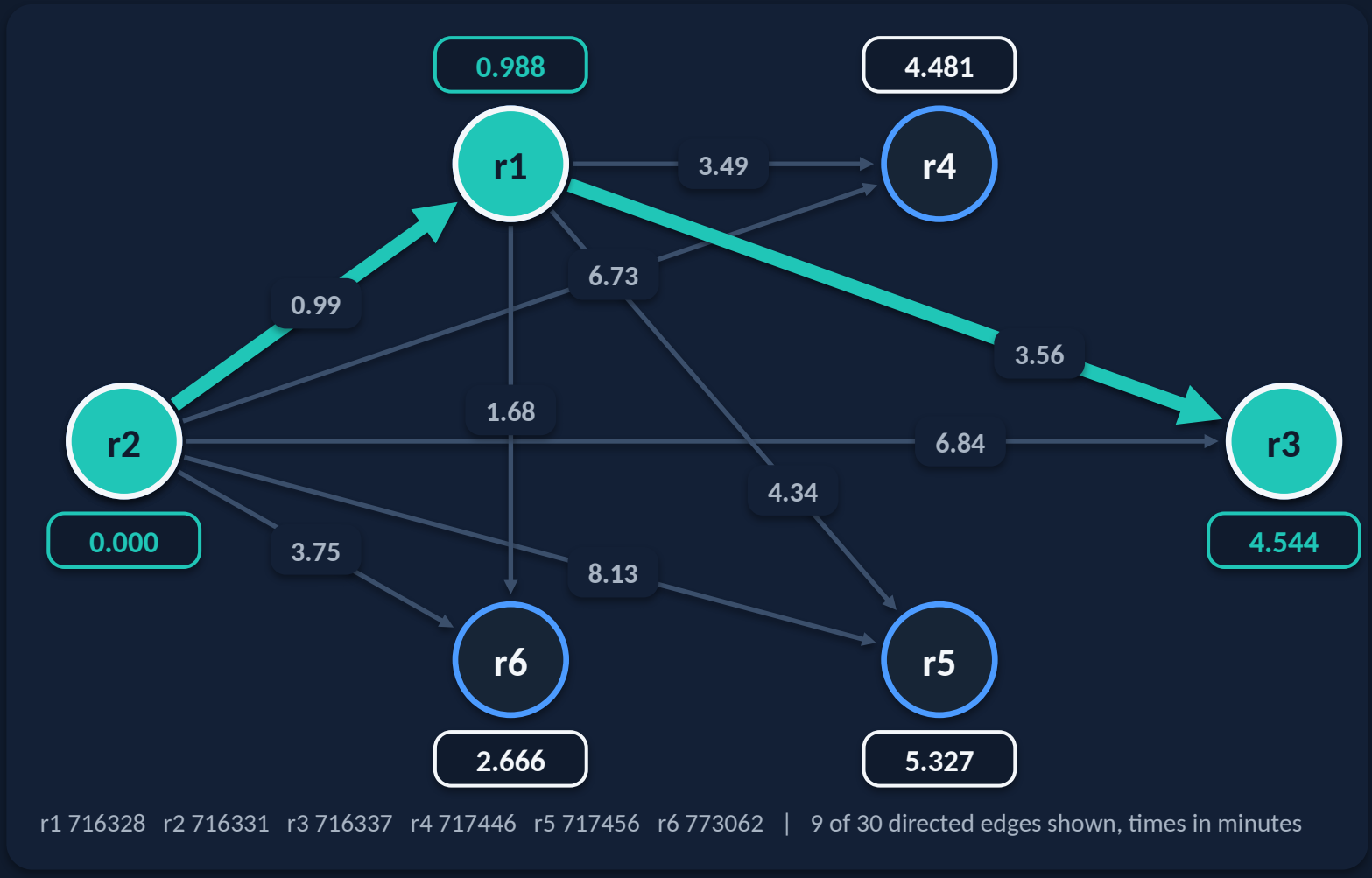
Sensor 717446 (r4): actual vs predicted speed, test period



LINEAR REGRESSION	PERSISTENCE BASELINE
MAE 4.86 | RMSE 6.93	MAE 5.14 | RMSE 7.54

Committed 2-day sample, held-out test split, all six sensors (522 predictions), mph. RMSE is 8.1% lower than persistence. This checks the pipeline; it is not a final accuracy claim.

Dijkstra: how the fastest route is chosen



r1 716328 r2 716331 r3 716337 r4 717446 r5 717456 r6 773062 | 9 of 30 directed edges shown, times in minutes

RESULT

Walk back from r3: r3, r1, r2. This is the recommended route.

SPEED TO TRAVEL TIME

minutes = metres / (mph × 26.8224)

r2 to r1: 874.5 / (33.01 × 26.8224) = **0.988**
r1 to r3: 5179.6 / (54.30 × 26.8224) = **3.557**

RECOMMENDED ROUTE

r2 > r1 > r3

ETA 4.544 min (direct 6.838)

The system runs: terminal demonstration

routing (SPL-1 command-line application)

```
$ routing --row 489 --traffic data/traffic_data.csv \
          --distances data/sensor_distances.csv

Where are you now?        (r1-r6 or sensor ID): r2
Where do you want to go?  (r1-r6 or sensor ID): r3

SPL-1 fastest route: Linear Regression forecast + Dijkstra
Forecast origin: row 489 (2012-03-02 22:45:00)
From / To      : 716331 -> 716337  (r2 -> r3)
Training the six Linear Regression models ...

Forecast speeds, 15 min after the origin:
  r1  716328   54.30 mph
  r2  716331   33.01 mph
  r3  716337   49.88 mph
  r4  717446   31.33 mph
  r5  717456   43.77 mph
  r6  773062   33.61 mph

Fastest route: 716331 -> 716328 -> 716337
  In route names: r2 -> r1 -> r3
  1  716331 -> 716328    874.5 m  33.01 mph  0.988 min
  2  716328 -> 716337   5179.6 m  54.30 mph  3.557 min
Total ETA: 4.544 minutes
```



- Plain C++17, standard library only
- Origin and destination: r1-r6 or sensor ID
- 426 of 426 software tests pass

Recreated from verified CLI output (committed sample, row 489). Not a screenshot.

Progress: what is built and tested

426 / 426

software tests passed across 8 test suites. These check code correctness, not prediction accuracy.

MERGED PULL REQUESTS

- #1 Data preparation
- #2 Predictive routing and CLI
- #3 Interactive origin and destination

COMPLETED	**Dataset acquisition and preprocessing**	`tools/prepare_data.cpp`
COMPLETED	**Data validation and cleaning**	`tools/validate_data.cpp`
COMPLETED	**Linear Regression baseline**	`src/linear_regression.cpp`
COMPLETED	**Model evaluation (MAE, RMSE, persistence)**	`baseline_report_15min.txt`
COMPLETED	**Dijkstra implementation**	`src/graph.cpp`
COMPLETED	**Predictive travel-time graph**	`src/travel_time_graph.cpp`
COMPLETED	**Route planning**	`src/route_planner.cpp`
COMPLETED	**Interactive CLI**	`src/routing_cli.cpp`
COMPLETED	**Origin and destination selection**	`r1-r6 or sensor ID`
COMPLETED	**Software testing**	`8 suites, 426 tests`

IN PROGRESS

Accuracy report on the full 34,272-row dataset; committed reports cover the 2-day sample

PLANNED

Neural Network, graphical interface, full system integration. Not implemented yet.

Roadmap: midterm to final

ALREADY COMPLETED

+ Data preparation and validation
+ Linear Regression speed prediction
+ Dijkstra shortest-time routing
+ Predictive travel-time graph
+ Interactive CLI
+ Origin / destination selection
+ Testing and integration of the command-line system

1

Neural Network

- Add a neural-network speed model
- Compare with the Linear Regression baseline
- Use suitable metrics (MAE, RMSE)

2

User Interface

- More intuitive graphical interface
- Easier sensor selection and route display
- Show predicted speeds and ETA clearly

3

System Integration

- Connect model, routing engine and UI
- Improve end-to-end usability
- Final testing, validation, refinement

PLANNED: none of these is implemented yet

From predicting traffic conditions to making smarter route decisions: building a more intelligent approach to urban mobility.

Thank you

Questions?